STALKS OF THE TENSOR PRODUCT OF SHEAVES OF MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The presheaf-level computation (F tensor_p O O')_x = F_x tensor_{O_x} O'_x (lemma-stalk-tensor-presheaf-modules). invoked at proof lines 2180-2182. Provides the stalk of the presheaf tensor product.
- Taking stalks commutes with sheafification. invoked at proof lines 2181-2182. Bridges the presheaf computation to the sheaf tensor product, which is the sheafification of the presheaf tensor product.

Source reference. Original lemma 007K: sources/stacks-sheaves/pinned-original-source.tex, source lines 1375–1375. Cited at proof lines 2180-2182 for the presheaf tensor stalk; load-bearing inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $\mathcal{O} \to \mathcal{O}'$ be a morphism of sheaves of rings on X. Let $\mathcal{F}$ be a sheaf $\mathcal{O}$-modules. Let $x \in X$. We have*

$$\mathcal{F}_x \otimes_{\mathcal{O}_x} \mathcal{O}'_x = (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{O}')_x$$

as $\mathcal{O}'_x$-modules.

Proof. Follows directly from Lemma 007K and the fact that taking stalks commutes with sheafification. □

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